

6.2 Shannon's Expansion Theorem

1 Circuit Decomposition

• Shannon's Expansion Theorem

- Can be used to decompose functions of large numbers of variables into functions of fewer variables.
- Offers a general decomposition technique for any function.
- Decomposing n -variable functions into $(n-1)$ -variable functions

$$Z(a, b, c, d, e, f)$$

$$= a' Z(0, b, c, d, e, f) + a Z(1, b, c, d, e, f)$$

$$= a' Z_0 + a Z_1$$

$$\text{ex. } Z = abcd'ef + a'b'c'def' + b'cd e'f$$

$$1^\circ \text{ Set } a = 0$$

$$Z = b'c'def' + b'cd e'f$$

$$2^\circ \text{ set } a = 1$$

$$Z_1 = bcd'e f' + b'cd e'f$$

$$3^o \quad Z = a' Z_0 + a Z_1$$

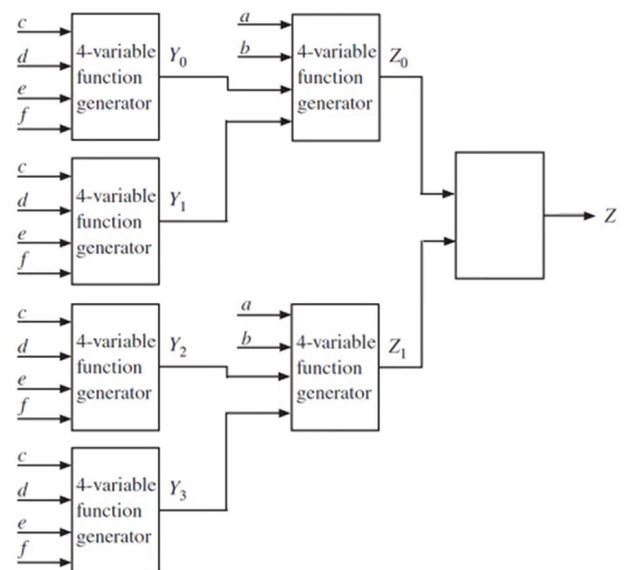
4' If only LUT4 is available, the
5-variable functions should be further
decomposed into 4-variable functions.

- Decomposing n -variable functions into
 $(n-2)$ -variable functions

Decomposing n -Variable Functions into $(n-2)$ -Variable Functions

- Realization of a general 6-variable function
using 4-variable functions:

$$\begin{aligned} & Z(a, b, c, d, e, f) \\ &= a'b' \cdot Z(0, 0, c, d, e, f) \\ &\quad + a'b \cdot Z(0, 1, c, d, e, f) \\ &\quad + ab' \cdot Z(1, 0, c, d, e, f) \\ &\quad + ab \cdot Z(1, 1, c, d, e, f) \\ &= \underline{a'b' \cdot Y_0 + a'b \cdot Y_1} \\ &\quad + \underline{ab' \cdot Y_2 + ab \cdot Y_3} \\ &= Z_0 + Z_1 \end{aligned}$$



Example: Realize a 6-variable function using 4-variable functions

- Consider the decomposition of the function:

$$Z = abcd'ef' + a'b'c'def' + b'cde'f$$

by applying the theorem around **a** and **b**:

– Substitute $a = b = 0 \Rightarrow Y_0 = c'def' + cde'f$

– Substitute $a = 0, b = 1 \Rightarrow Y_1 = 0$

– Substitute $a = 1, b = 0 \Rightarrow Y_2 = cde'f$

– Substitute $a = b = 1 \Rightarrow Y_3 = cd'ef'$

$$\Rightarrow Z = a'b' \cdot Y_0 + ab' \cdot Y_2 + ab \cdot Y_3$$

$\Rightarrow$ Five 4-variable function generators will be sufficient to implement this function. (next page)

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Decomposing n -Variable Functions into $(n-2)$ -Variable Functions

- Realization of a general 6-variable function using 4-variable functions:

$$Z(a, b, c, d, e, f)$$

$$= a'b' \cdot Z(0, 0, c, d, e, f)$$

$$+ a'b \cdot Z(0, 1, c, d, e, f)$$

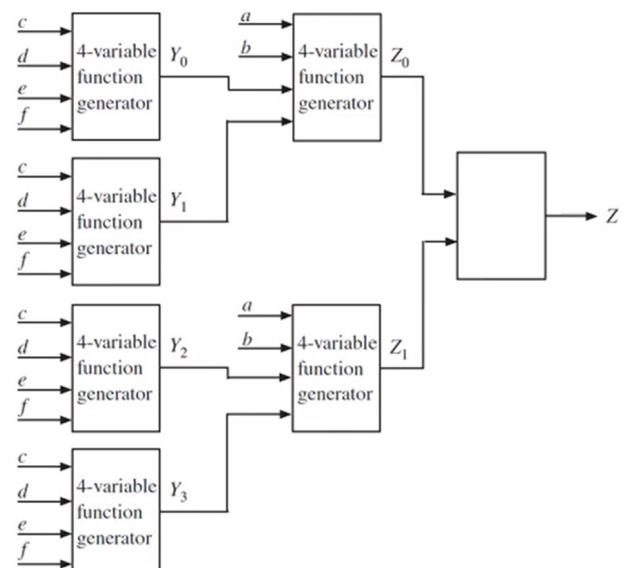
$$+ ab' \cdot Z(1, 0, c, d, e, f)$$

$$+ ab \cdot Z(1, 1, c, d, e, f)$$

$$= a'b' \cdot Y_0 + a'b \cdot Y_1$$

$$+ ab' \cdot Y_2 + ab \cdot Y_3$$

$$= Z_0 + Z_1$$



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Brief Summary

■ Shannon's decomposition:

- May decompose an n -variable function into
 - 2 $n-1$ variable functions and a 2-to-1 MUX or
 - 4 $n-2$ variable functions and a 4-to-1 MUX or ...
- As the # of variables (n) increases, the # of LUTs required to realize an n -variable function increases rapidly.
- It is very inefficient to realize MUXs using LUTs.
 - ⇒ The availability of MUXs can greatly reduce the # of LUTs needed.
 - ⇒ Some FPGAs provide **MUXs** in addition to **LUT4s** (e.g., the Xilinx Spartan FPGA).

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Example

- ### ■ Implement a 7-variable function using 4-input LUTs and 2-to-1 MUXs.

<Ans.>

- 7-variable function generator = two 6-variable function generators + a 2-to-1 MUX ... (i)
- ✓ 6-variable function generator = two 5-variable function generators + a 2-to-1 MUX ... (ii)
- ✓ 5-variable function generator = two 4-variable function generators + a 2-to-1 MUX ... (iii)

Substituting (iii) into (ii)

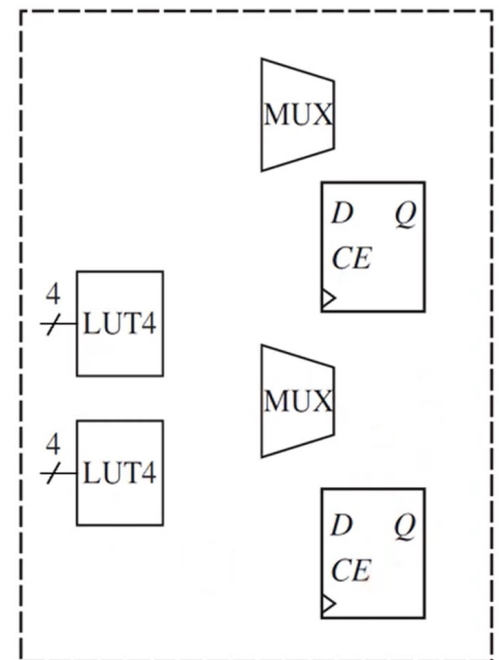
⇒ 6-variable function generator = Four 4-variable function generators + three 2-to-1 MUXes ... (iv)

Substituting (iv) into (i)

⇒ 7-variable function generator = Eight 4-variable function generators + seven 2-to-1 MUXes

Xilinx Spartan FPGA

- The Xilinx Spartan FPGA:
 - provides *MUXs* in addition to the general *4-variable LUTs*.
 - A logic unit in these FPGAs is called a *slice*.
 - Contains two 4-input LUTs, two 2-to-1 MUXs, two D flip-flops, ...



A Xilinx Spartan *slice*

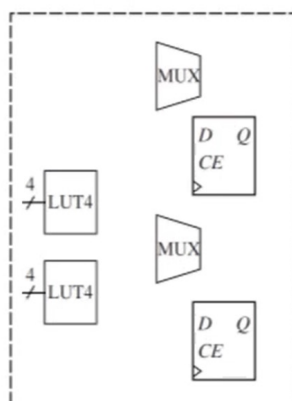
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Example

- Implement a 7-variable function using Xilinx Spartan slices.

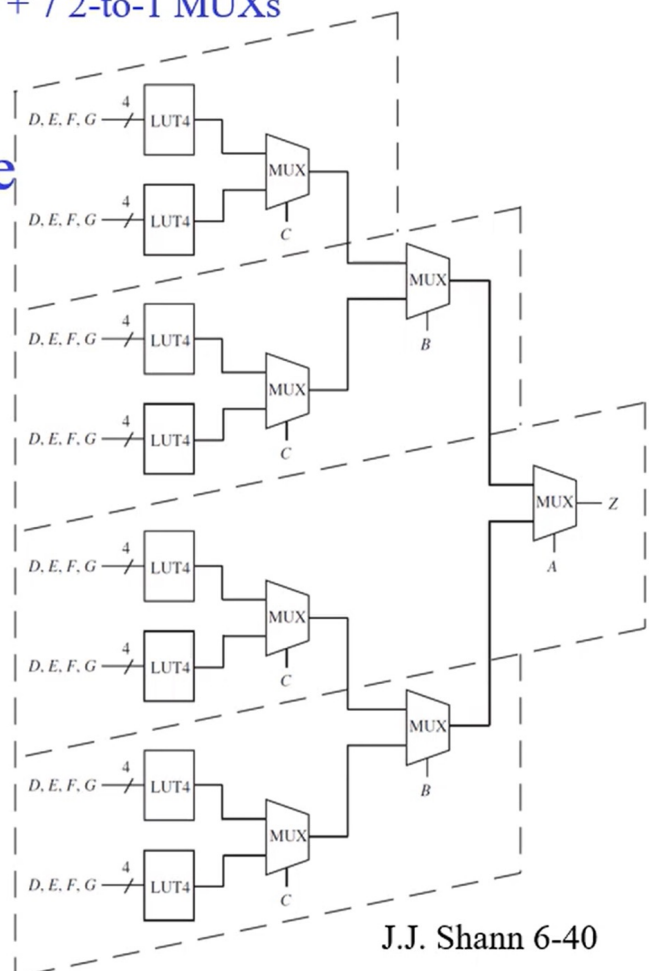
<Ans.>

Four slices are required.



A Xilinx Spartan *slice*

8 4-variable function generators
+ 7 2-to-1 MUXs



J.J. Shann 6-40